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Indian Institute of Information Technology and Management Gwalior

Computer Graphics Lab :
07

Spline-based Object Representation

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Teaching Assistants

(TAs)

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Objective of this lab

The objective of this lab is to represent and approximate 2D object boundaries using spline curves. The lab involves selecting control points and studying first-order, second-order, and third-order splines for smooth and accurate object representation.

Demo Assignment – 01

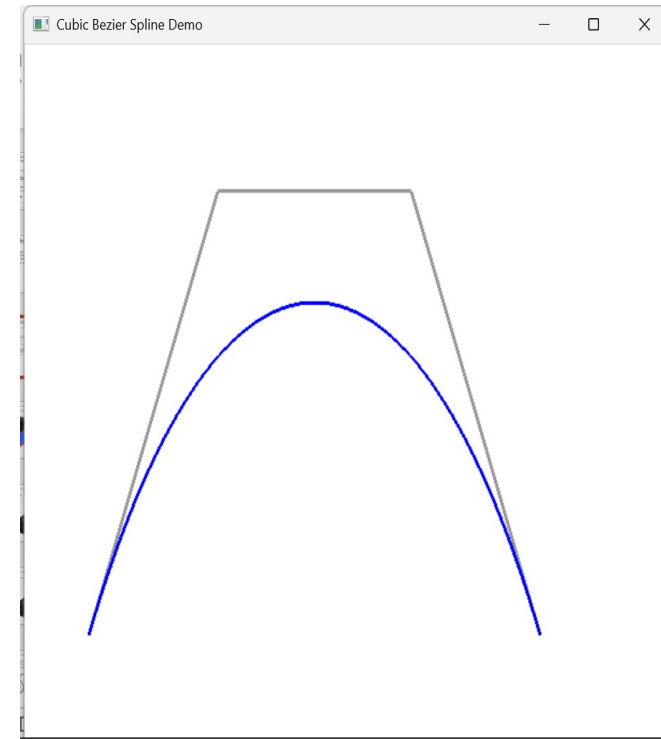
Draw a cubic Bézier spline curve using four control points and display the resulting curve on the computer screen using OpenGL.

```
#include <GL/freeglut.h>
void display()
{
    glClear(GL_COLOR_BUFFER_BIT);

    float x0 = 1.0f, y0 = 2.0f;
    float x1 = 3.0f, y1 = 8.0f;
    float x2 = 6.0f, y2 = 8.0f;
    float x3 = 8.0f, y3 = 2.0f;
    // Control polygon
    glColor3f(0.6f, 0.6f, 0.6f);
    glBegin(GL_LINE_STRIP);
        glVertex2f(x0, y0);
        glVertex2f(x1, y1);
        glVertex2f(x2, y2);
        glVertex2f(x3, y3);
    glEnd();
    // Cubic Bezier curve
    glColor3f(0.0f, 0.0f, 1.0f);
    glLineWidth(3.0f);
    glBegin(GL_LINE_STRIP);
    for (int i = 0; i <= 100; i++)
    {
        float t = i / 100.0f;
        float u = 1.0f - t;
        float x = u*u*u*x0
            + 3*u*u*t*x1
            + 3*u*t*t*x2
            + t*t*t*x3;
        float y = u*u*u*y0
            + 3*u*u*t*y1
            + 3*u*t*t*y2
            + t*t*t*y3;
        glVertex2f(x, y);
    }
    glEnd();
    glFlush();
}

void init()
{
    glClearColor(1.0f, 1.0f, 1.0f, 1.0f);
    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    glOrtho(0, 10, 0, 10, -1, 1);
}

int main(int argc, char** argv)
{
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_SINGLE |
        GLUT_RGB);
    glutInitWindowSize(600, 600);
    glutCreateWindow("Cubic Bezier Spline
        Demo");
    init();
    glutDisplayFunc(display);
    glutMainLoop();
    return 0;
}
```



Assignment - 02

- ❖ **Question -02** : Using OpenGL, localize 3D control points on the object shown below.



Assignment - 03

❖ **Question -03** : Approximate the boundary using:

- First order spline
- 2nd Order spline
- 3rd order spline



Assignment - 04

❖ **Question -04** : Generate 1000 control points and draw the boundary again using:

- First order spline
- 2nd Order spline
- 3rd order spline

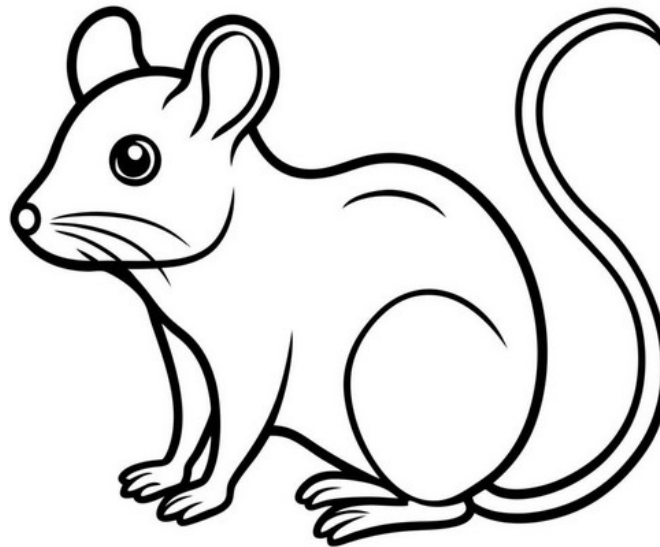


Assignment - 05

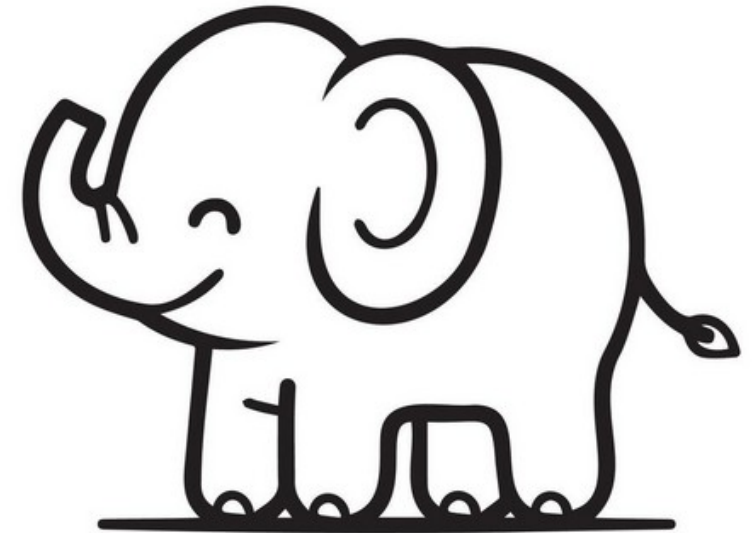
- ❖ **Question -05** : Indicate suitable (set of control points) no. of control points (say N) on below objects and approximate them using 3rd order spline.
- ❖ **Question -06**: Generate another N no of control points and repeat the experiment.



(a)
)



(b)
)



(c)
)

Thank you !